
1. Boxes

Program Name: Boxes.java

Input File: boxes.dat

Judges Input File

```
12
2
1x2x3 2x3x4
2
3x1x2 2x3x4
2
1x2x3 2x3x3
3
1x2x3 2x3x4 3x4x4
5
1x2x3 4x3x2 4x3x6 4x4x7 10x10x10
3
10x10x10 1x1x1 2x2x2
2
1x2x3 2x4x3
2
1x2x3 3x4x2
2
1x2x3 4x3x2
2
1x2x3 3x2x4
2
1x2x3 2x3x4
2
1x2x3 4x2x3
```

Judges Output to Screen

```
1
1
0
1
2
0
1
1
1
1
1
1
1
```

2. Encryption

Program Name: Encryption.java

Input File: encryption.dat

Judges Input File

```
8
1234567890123456
4536278290123476
7685438920456431
5647839201124576
8793333456111123
1111111111111111
0987654321123456
7777777777777777
```

Judges Output to Screen

```
0189452389672301
0312599489674301
5243561012970831
1423695089784312
6054010088239088
8888888888888888
5476103289982301
4444444444444444
```

3. FizzBang!!!

Program Name: Fizz.java

Input File: fizz.dat

Judges Input File

```
8
5 5
6 10
10 15
20 30
1 2
150 155
1999999995 2000000000
1111111 1111114
```

Judges Output to Screen (see next page)

3. FizzBang!!! (cont.)

Judges Output to Screen

```
Data Set 1
Bang
Data Set 2
Fizz
7
8
Fizz
Bang
Data Set 3
Bang
11
Fizz
13
14
FizzBang
Data Set 4
Bang
Fizz
22
23
Fizz
Bang
26
Fizz
28
29
FizzBang
Data Set 5
1
2
Data Set 6
FizzBang
151
152
Fizz
154
Bang
Data Set 7
FizzBang
1999999996
1999999997
Fizz
1999999999
Bang
Data Set 8
1111111
1111112
Fizz
1111114
```

4. Intersection

Program Name: Intersection.java

Input File: intersection.dat

Judges Input File

```
8
3
1 2 3 4 5 5
3 4 5 6 7 5
4 5 1 2 3 5 6 7 8 9
5
10 11 15 61 42 14 112 34
12 13 11 61 42 10 34 12 14 112
10 10 13 11 112 54 61 13 13 34
11 112 15 42 13 34 17 10
71 10 11 112 75 63 15 42 34 56 67
4
1 2
3 4 5
6 7 8 9
10 11 12 13 14
2
1 11 111 1111 11111
1 11 111 1111 11111
3
1 2 3
1 2 3
1 2 3
5
100 150 200 250 300 350
1 2 3 4
100 150 200 250
1 2 3 4
300 400
7
1 1 1 4 5 5 7 8 8 9 10 11 14 15 16 17 20
11 10 14 15 4 5 3 1
14 15 4 3 3 1
14 15 4 3 8 1 12 11
11 10 12 123 23 24 14 15 4 5 3 1
14 15 4 14 3 121 76 1
14 15 4 3 8 1 12 11 40 21
10
45 46 4 47 48
55 56 4 57 58 59
1 2 4
3 4
5 4 6
4 7 8
4 9 10
1 2 4
1 2 4
3 4
```

Judges Output to Screen (see next page)

4. Intersection (cont.)

Judges Output to Screen

```
3 4 5
10 11 34 112
NO COMMON ELEMENT FOUND
1 11 111 1111 11111
1 2 3
NO COMMON ELEMENT FOUND
1 4 14 15
4
```

5. A Long Walk on Elm Street

Program Name: LongWalk.java

Input File: longwalk.dat

Judges Input File

```
10
Elm Street
3
1001
1210
514
El Camino Real
4
99999
12
10000
9999
Broadway
3
1
999999
123456
University Avenue
5
960
844
414
895
948
Copperas
Cove

8
199
97
167
149
166
59
154
1
Great James Place
5
708023
732133
2601
711632
755582
9th
Street

7
169
22
415
```

208
109
245
202
Tiny Blvd.
2
1

5. A Long Walk on Elm Street (cont.)

2
Great Big Honking Street
2
999999
1
Great Big Honking Avenue
2
999998
999997

Judges Output to Screen

The two houses farthest apart on Elm Street are
514 Elm Street and 1210 Elm Street
The two houses farthest apart on El Camino Real are
12 El Camino Real and 99999 El Camino Real
The two houses farthest apart on Broadway are
1 Broadway and 999999 Broadway
The two houses farthest apart on University Avenue are
414 University Avenue and 960 University Avenue
The two houses farthest apart on Copperas Cove are
1 Copperas Cove and 199 Copperas Cove
The two houses farthest apart on Great James Place are
2601 Great James Place and 755582 Great James Place
The two houses farthest apart on 9th Street are
22 9th Street and 415 9th Street
The two houses farthest apart on Tiny Blvd. are
1 Tiny Blvd. and 2 Tiny Blvd.
The two houses farthest apart on Great Big Honking Street are
1 Great Big Honking Street and 999999 Great Big Honking Street
The two houses farthest apart on Great Big Honking Avenue are
999997 Great Big Honking Avenue and 999998 Great Big Honking Avenue

6. Mailboxes, Etc.

Program Name: Mail.java

Input File: mail.dat

Judges Input File

```
9
3
0
2 1 3
0
3
1 1
1 2
0
3
1 2
1 3
0
3
0
0
0
0
3
1 3
1 2
1 1
1
1 1
30
0
1 1
1 2
1 3
1 4
1 5
1 6
1 7
1 8
1 9
1 10
1 11
1 12
1 13
1 14
1 15
1 16
1 17
1 18
1 19
1 20
1 21
1 22
1 23
1 24
1 25
1 26
1 27
```

6. Mailboxes, Etc. (cont.)

```
1 28
1 29
5
5 1 2 3 4 5
5 1 2 3 4 5
5 1 2 3 4 5
5 1 2 3 4 5
5 1 2 3 4 5
8
2 3 4
1 5
1 3
1 4
3 6 7 8
0
0
0
```

Judges Output to Screen

```
3
1
3
1
2
1
30
5
5
```

7. Mathabetical Order

Program Name: Mathabet.java

Input File: mathabet.dat

Judges Input File

13
ag
bury
cry
dad
asap
b
c
d
hi
jj
stow
sty
crw

Judges Output to Screen

crw
cry
sty
bury
stow
d
dad
jj
asap
c
ag
hi
b

Word	Resulting Number	Prime	Even
crw	31823	N	N
cry	31825	N	N
sty	192025	N	N
bury	2211825	N	N
stow	19201523	N	N
d	4	N	Y
dad	414	N	Y
jj	1010	N	Y
asap	119116	N	Y
c	3	Y	N
ag	17	Y	N
hi	89	Y	N
b	2	Y	Y

8. Room for Improvement

Program Name: Room.java

Input File: room.dat

Judges Input File

```
7
5 5
#####
#...#
#.@.#
#...#
#####
7 19
#####
#.....#
#.....#
#...###.....#..
#.@.#.....#..
#...###.....#..
#####
3 3
###
#@#
###
7 7
#####
#.....#
#...###.#
#.#@#.#
#...###.#
#.....#
#####
7 7
#####
#.....#
#...###.#
#.#@#.#
#...###.#
#.....#
#####
8 7
#####
#.....#
#####.#
#...###.#
#.#@#.#
#...###.#
#.....#
#####
11 13
#####
#.....#.#
#.....#.#
#.#.....#.#.#
#...###.#.#.#
#...###.#@#.#.#
#...###.#.#.#
```

8. Room for Improvement (cont.)

###.....##
#####.
#.....#
#####

Judges Output to Screen

\$9
\$27
\$1
\$1
\$18
\$20
\$29

Program Name: Stairs.java **Input File: none**

Judges Input File

There is no input for this problem.

Judges Output to Screen

[illegible]

10. Top Ten Reasons ...

Program Name: TopTen.java **Input File:** topten.dat

Judges Input File

1. Taking hardware home to show my parents.
2. Eating junk food for 2 days.
3. Staying overnight in a crowded hotel room.
4. Riding miles and miles in a crowded vehicle.
5. Staying up all night to play video games.
6. Listening to loud music on my ipod.
7. Getting up early in the morning.
8. Getting to miss 2 days of school.
9. Meeting new friends.
10. Learning about Computer Science.

Judges Output to Screen

10. Learning about Computer Science.
9. Meeting new friends.
8. Getting to miss 2 days of school.
7. Getting up early in the morning.
6. Listening to loud music on my ipod.
5. Staying up all night to play video games.
4. Riding miles and miles in a crowded vehicle.
3. Staying overnight in a crowded hotel room.
2. Eating junk food for 2 days.
1. Taking hardware home to show my parents.

11. Vector Addition

Program Name: Vectors.java **Input File:** vectors.dat

Judges Input File

```
6
(3,4) (-1,3)
(5,2) (-6,0) (6,-2)
(19,12) (4,-2) (5,6) (1,4)
(1,1) (0,0) (21,-1)
(-3,-4) (-5,-4) (-8,-9) (-7,-6) (-5,-6) (-4,-7)
(-1,2) (2,-8) (-4,0) (0,-8)
```

Judges Output

```
(2,7)
(5,0)
(29,20)
(22,0)
(-32,-36)
(-3,-14)
```

12. Widget Wackiness

Program Name: Widget.java

Input File: widget.dat

Judges Input File

```
9
10 2 5
7 1 4
56 2 4
0 1 2
16 3 3
16 3 4
64 1 2
1 2 2
2 1 3
```

Judges Output to Screen

```
6
9
37
0
7
6
127
0
2
```